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%HW\_2\_2 Nouh Shaikh

clear; clc; close all;

**a)**

T\_plot = linspace(0, 1200, 1000);

c0 = 0.99403;

c1 = 1.671e-4;

c2 = 9.7215e-8;

c3 = -9.5838e-11;

c4 = 1.9520e-14;

cp\_plot = c0 + c1\*T\_plot + c2\*T\_plot.^2 + c3\*T\_plot.^3 + c4\*T\_plot.^4;

figure;

plot(T\_plot, cp\_plot, 'b-', 'LineWidth', 2);

hold on;

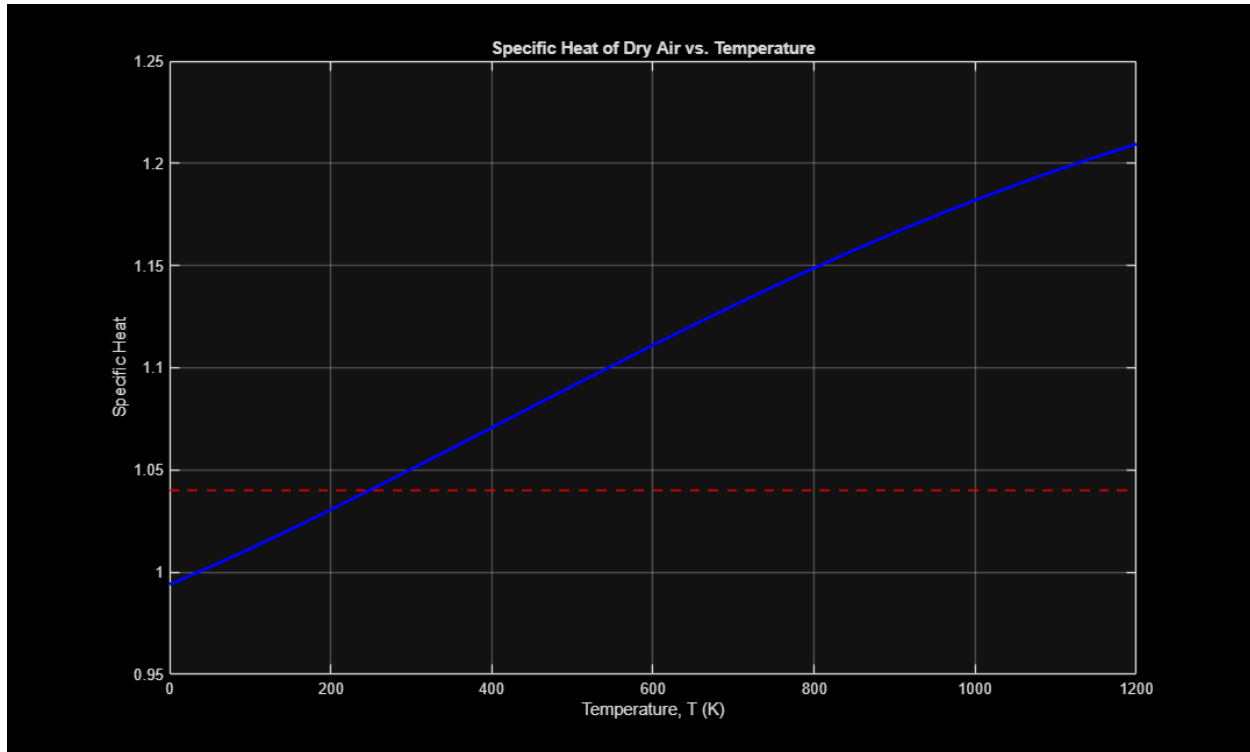
ylines(1.04, 'r--', 'LineWidth', 1.5);

grid on;

xlabel('Temperature, T (K)');

ylabel('Specific Heat')

title('Specific Heat of Dry Air vs. Temperature');



**b)**

```
g = @(T) (c0 + c1*T + c2*T.^2 + c3*T.^3 + c4*T.^4) - 1.04;
```

```
xL = 0;
xU = 300;
```

```
if g(xL) * g(xU) > 0
    error('The chosen initial bracket does not contain a root.');
```

```
end
```

```
fprintf('Valid initial bracket found: [xL, xU] = [%.1f, %.1f]\n', xL, xU);
```

```
Valid initial bracket found: [xL, xU] = [0.0, 300.0]
```

**c)**

```
tol = 1e-5;
max_iter = 100;
iter = 0;
Ea = inf;
xr_prev = 0;
```

```
fprintf('\nIter\t xL\t\t xU\t\t xr\t\t Ea\n');
```

```
fprintf('-----\n');
```

---

```

while Ea > tol && iter < max_iter

    xr = (xL + xU) / 2;
    iter = iter + 1;

    if iter == 1
        Ea = inf;
    else
        Ea = abs(xr - xr_prev);
    end

    if Ea == inf
        fprintf('%2d\t %.4f\t %.4f\t %.4f\t      inf\n', iter, xL, xU, xr);
    else
        fprintf('%2d\t %.4f\t %.4f\t %.4f\t %.8f\n', iter, xL, xU, xr, Ea);
    end

    if g(xL) * g(xr) < 0
        xU = xr;
    else
        xL = xr;
    end

    xr_prev = xr;
end

% solved point on the graph
plot(xr, 1.04, 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r');
fprintf('\n-----\n');
fprintf('Final Temperature for cp = 1.04 kJ/(kg K): %.4f K\n', xr);
fprintf('Total Iterations Completed: %d\n', iter);

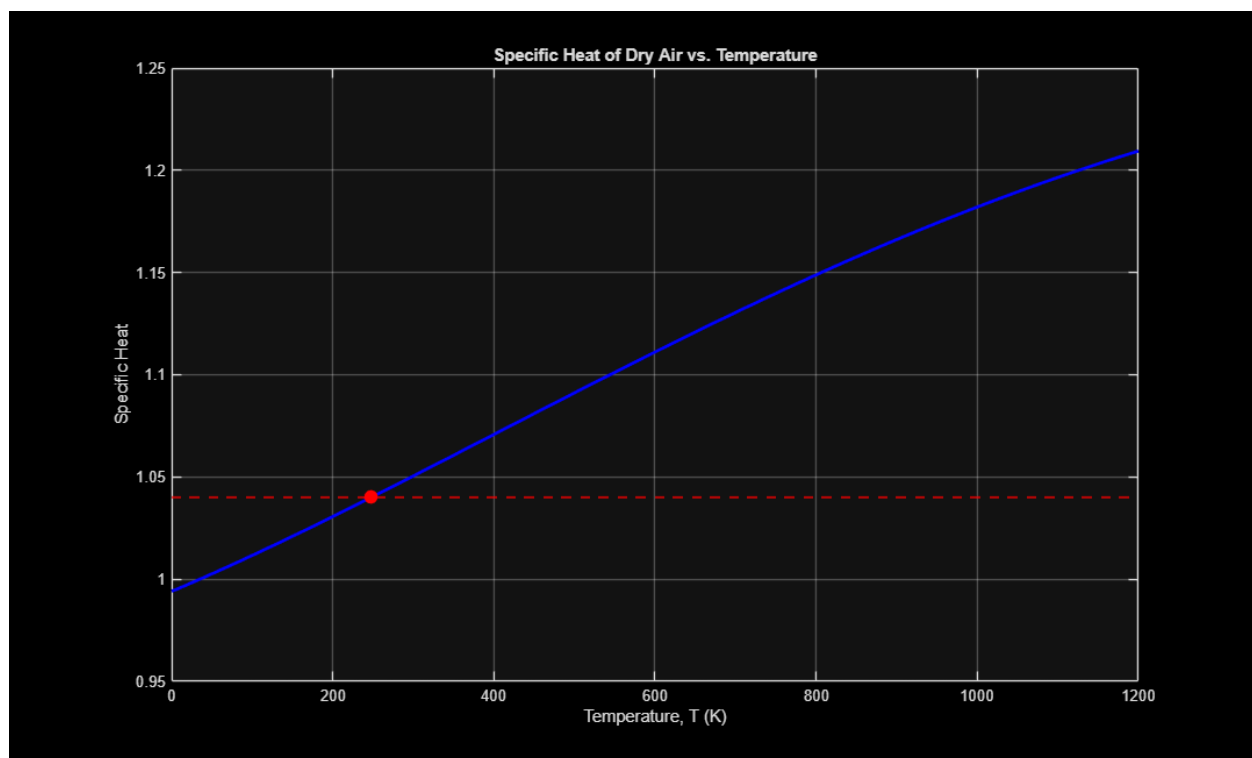
```

| <i>Iter</i> | <i>xL</i> | <i>xU</i> | <i>xr</i> | <i>Ea</i>   |
|-------------|-----------|-----------|-----------|-------------|
| 1           | 0.0000    | 300.0000  | 150.0000  | inf         |
| 2           | 150.0000  | 300.0000  | 225.0000  | 75.00000000 |
| 3           | 225.0000  | 300.0000  | 262.5000  | 37.50000000 |
| 4           | 225.0000  | 262.5000  | 243.7500  | 18.75000000 |
| 5           | 243.7500  | 262.5000  | 253.1250  | 9.37500000  |
| 6           | 243.7500  | 253.1250  | 248.4375  | 4.68750000  |
| 7           | 243.7500  | 248.4375  | 246.0938  | 2.34375000  |
| 8           | 246.0938  | 248.4375  | 247.2656  | 1.17187500  |
| 9           | 247.2656  | 248.4375  | 247.8516  | 0.58593750  |
| 10          | 247.2656  | 247.8516  | 247.5586  | 0.29296875  |
| 11          | 247.5586  | 247.8516  | 247.7051  | 0.14648438  |
| 12          | 247.5586  | 247.7051  | 247.6318  | 0.07324219  |
| 13          | 247.6318  | 247.7051  | 247.6685  | 0.03662109  |
| 14          | 247.6685  | 247.7051  | 247.6868  | 0.01831055  |
| 15          | 247.6868  | 247.7051  | 247.6959  | 0.00915527  |

---

|    |          |          |          |            |
|----|----------|----------|----------|------------|
| 16 | 247.6868 | 247.6959 | 247.6913 | 0.00457764 |
| 17 | 247.6868 | 247.6913 | 247.6891 | 0.00228882 |
| 18 | 247.6868 | 247.6891 | 247.6879 | 0.00114441 |
| 19 | 247.6879 | 247.6891 | 247.6885 | 0.00057220 |
| 20 | 247.6885 | 247.6891 | 247.6888 | 0.00028610 |
| 21 | 247.6885 | 247.6888 | 247.6886 | 0.00014305 |
| 22 | 247.6885 | 247.6886 | 247.6886 | 0.00007153 |
| 23 | 247.6885 | 247.6886 | 247.6885 | 0.00003576 |
| 24 | 247.6885 | 247.6885 | 247.6885 | 0.00001788 |
| 25 | 247.6885 | 247.6885 | 247.6885 | 0.00000894 |

-----  
Final Temperature for  $c_p = 1.04 \text{ kJ}/(\text{kg K})$ : 247.6885 K  
Total Iterations Completed: 25



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